

Titanic Dataset Exploratory Data Analysis

Introduction

This report presents an Exploratory Data Analysis (EDA) of the Titanic dataset. The goal is to understand passenger characteristics, identify patterns related to survival, and derive meaningful insights using statistical summaries and visual analysis.

Dataset Overview

The Titanic dataset contains passenger information such as age, gender, passenger class, fare, embarkation point, and survival status. It is widely used for data analysis and machine learning projects.

Data Cleaning

Data cleaning involved checking for missing values, duplicate records, and inconsistent data types. Missing values in Age and Embarked can be handled using median and mode imputation techniques.

Univariate Analysis

Histograms and count plots were used to analyze individual variables. Most passengers were between 20 and 40 years old, and the majority belonged to the third passenger class.

Bivariate Analysis

Relationships between variables were explored using count plots, box plots, and scatter plots. Female passengers showed a higher survival rate than male passengers. First-class passengers also had a better chance of survival.

Correlation Analysis

A correlation heatmap was used to examine relationships among numerical variables. Fare showed a positive relationship with survival, while passenger class showed a negative relationship.

Outlier Detection

Boxplots were used to identify outliers. Significant outliers were observed in the Fare column, indicating that some passengers paid substantially higher ticket prices.

Key Insights

1. Female passengers had higher survival rates.
2. First-class passengers were more likely to survive.
3. Fare had a positive impact on survival probability.
4. Age showed a weaker relationship with survival compared to gender and class.
5. The dataset contained missing values and fare outliers that required attention.

Conclusion

The Titanic EDA revealed clear patterns affecting passenger survival. Gender, passenger class, and fare were the most influential factors. Exploratory analysis provided valuable insights and prepared the dataset for future predictive modeling.